



MINING DEALER BEST PRACTICE

Improved Procedure for Removing the Camshaft from D11 Fusion Tractors

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

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December
1.10-241218-1

1.0 Introduction

Removing the camshaft from the C32 engines of D11 fusion tractors has always been a time-consuming and high-risk activity due to the movement of heavy parts, making it costly and dangerous. To replace the camshaft, it is necessary to remove the implement pumps or the fan and front mask of the equipment.

The need to remove or tilt the front mask with the engine radiator further increases the complexity and dangers associated with this activity. By exploring new strategies and techniques for camshaft removal, we aim to not only increase operational efficiency but also ensure the safety of technicians, minimize costs associated with the activity, and eliminate the need to move the radiator.

After the maintenance and repair of D11 fusion tractors through the implementation of this procedure, it was found that there was not only a reduction in risks but also costs with labor and machine downtime.

This procedure will present the removal of the camshaft, identifying its points and associated challenges. Finally, the expected benefits of this approach include increased safety for the technicians involved, a significant reduction in maintenance costs, and a significant increase in the efficiency of maintenance operations for tractors of the D11 KSN series.

2.0 Best Practice Description

Current procedure for removing the LD valve camshaft.

Tools:

Required Tools			
Tool	Part Number	Part Description	Qty
A	9U-7257	Cradle As	3
B	9U-7240	Camshaft Hook	2

Steps:

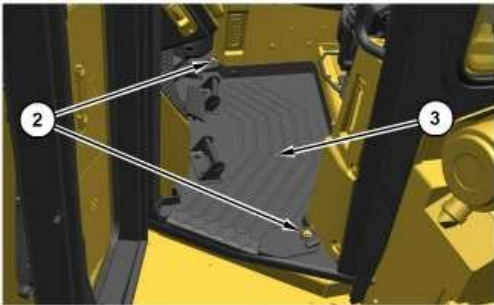
1. Prepare the equipment for maintenance using the procedure Prepare the Machine for Maintenance (M0082082-10)
2. Use the procedure System Pressure Release (M0082082-10) to depressurize all systems of the machine.

Note:

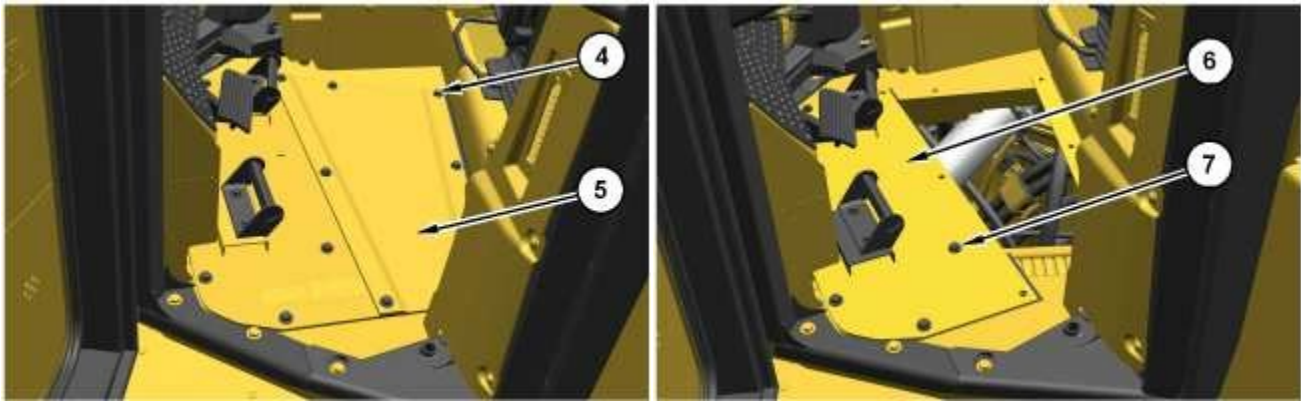
For safety reasons, this following process should only be carried out when the machine is cold condition.

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3. Remove the bolts (2) and (3) from the protection of the basement below the cabin carpet.



4. Remove the bolts (4) and (7), then remove the protections (5) and (6).



5. Remove the valve cover.

NOTICE
Keep all parts clean from contaminants.

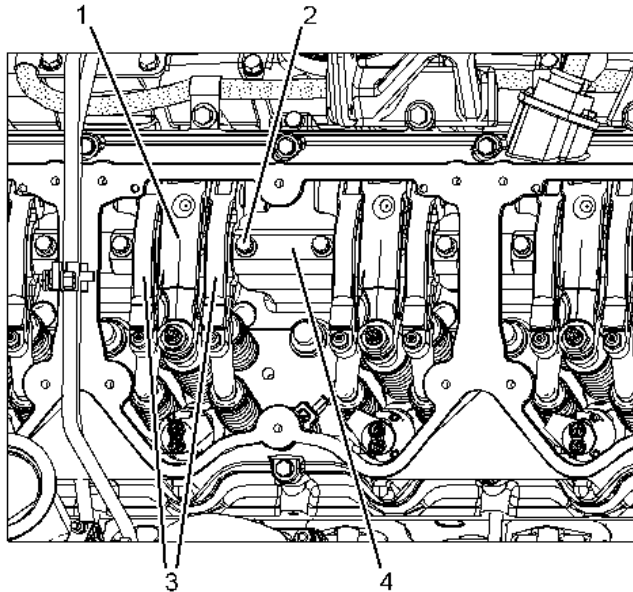
Contaminants may cause rapid wear and shortened component life.

NOTICE
Do not turn the crankshaft or the camshaft while the camshaft gear is removed. If the rear gear group is not correctly timed during installation, interference can occur between the pistons and the valves, resulting in damage to the engine.

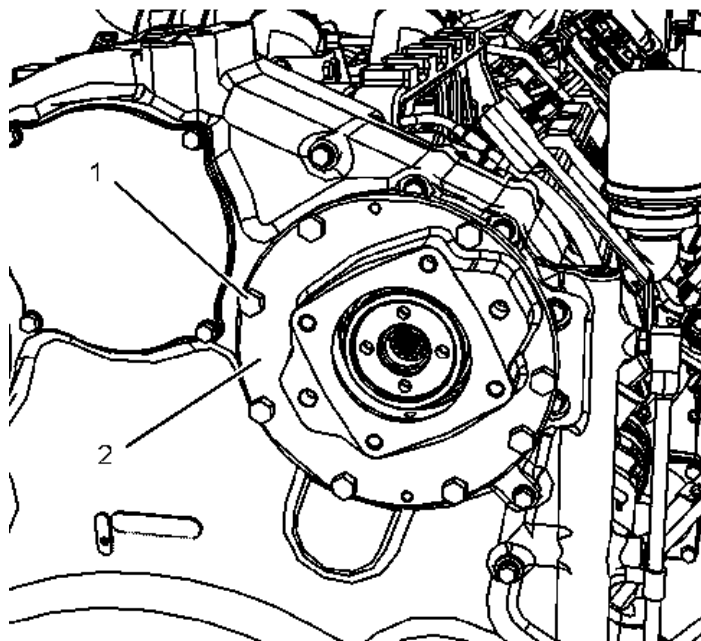
NOTICE
Care must be used when removing or installing the camshaft. Do not damage the finished surfaces of the camshaft or the camshaft bearings.

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6. Remove the bolts (2) to remove the rocker arm assemblies.



7. Drain the hydraulic oil using the procedure Hydraulic System Oil - Change (M0082082-10) and the ELC using the procedure Cooling System Coolant (ELC) - Change (M0082082-10).
 8. Remove the implement pump using the procedure Piston Pump (Implement) - Remove and Install (M0095201-06)
 9. Remove the fan pump using the procedure Hydraulic System Oil - Change (M0082082-10).
 10. Remove the transmission pump using the procedure Gear Pump (Power Train) - Remove and Install (M0095202-07) if the shaft to be removed is the left one.
 11. Remove the bolts 1 to remove the pump flange.



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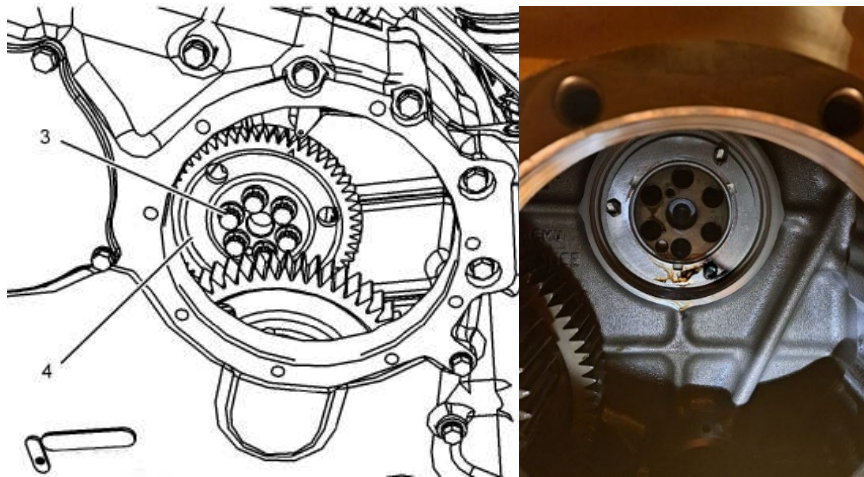
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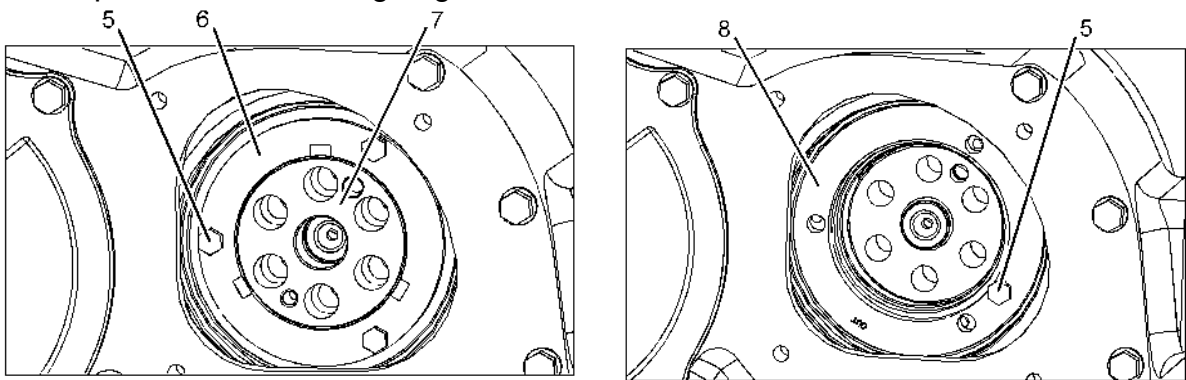
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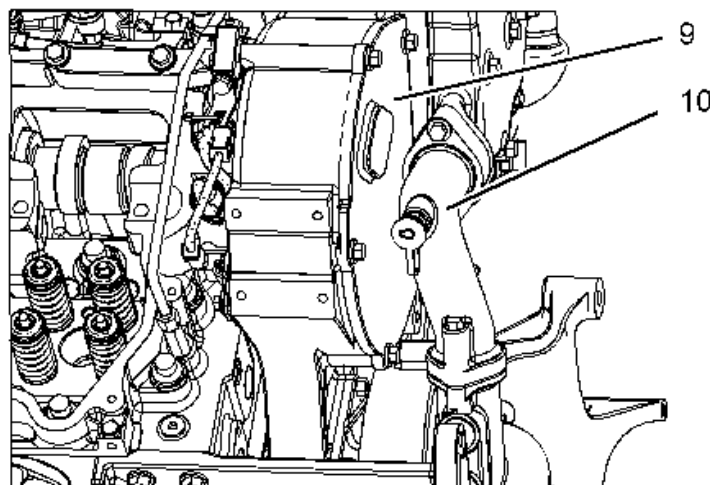
12. Put the engine at the point and check the alignment of the marks on the rear gear and lock the assembly. Remove the bolts (3) and the gear (4).



13. Remove the bolts (5) and the adapter (7) to remove the plate (6). Use the bolt (5) to help remove the plate and the sealing rings.



14. Remove the cover and the tubing.



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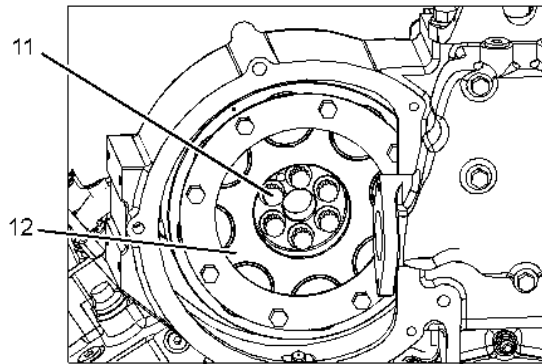
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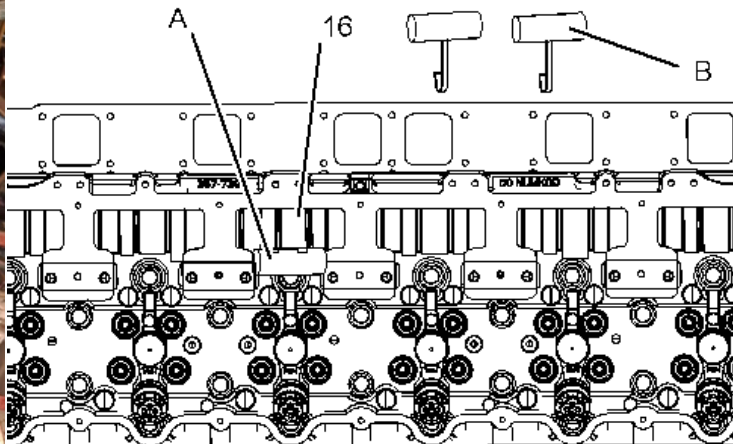
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15. Remove the bolts from the front gear (it is not necessary to remove the front gear from the housing). It is also not necessary to remove the thrust washer.



16. Use tool B to move the shaft, install two bolts on the shaft to help move it.



17. Using tool B to carefully slide the camshaft out of the cylinder head. This will prevent the camshaft from binding in the camshaft bearings.



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3.0 Implementation Steps

After the release of service letter PS47495 for the D11 KSN, it was identified that there was an urgent need to optimize the repair process in order to minimize equipment downtime. In June 2023, measurements, and evaluations of the new configuration of the machine's basement components were carried out. This included the relocation of groups and pumps in the new series, ensuring enough space for the shaft removal through the rear of the engine. In order to speed up the process and ensure quick availability of the equipment, the procedure was implemented as planned, resulting in a 2/3 reduction in repair time. The collaboration between engineering and field technicians was fundamental to develop and implement this improvement. Due to the nature of the adaptation of the procedures, this implementation can be easily replicated in other dealerships, as it is a process re-adaptation that does not require additional tools.

4.0 Benefits

1. Increased Safety:
 - Eliminates the step of tilting or removing the radiator guard, reducing the risk of accidents and injuries for technicians.
2. Reduced Machine Downtime:
 - Repair time for camshaft removal is reduced by 2/3, from 80 to 24 hours.
 - This means that the machine is available for operation more quickly, increasing productivity and efficiency.
3. Reduced Labor Costs:
 - The reduction in repair time leads to a 2/3 reduction in labor costs.
 - This means that the company saves money on each repair, which can lead to significant savings over time.

5.0 Resources Required

Required Tools			
Tool	Part Number	Part Description	Qty
A	9U-7257	Cradle As	3
B	9U-7240	Camshaft Hook	2

6.0 Supporting Attachments / References

REN9217-61_i02319602
REN9217-61_i02319611

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7.0 Related Best Practices

N/A

8.0 Acknowledgements

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